

Hien Ohnaka

Engineer

✉ ohnaka.hien@gmail.com 📍 Japan 🌐 i17oonaka-h ✖ nehi_h

Profile

Full-time engineer at LY Corporation, Japan.

Experience

- | | |
|--|---|
| LY Corporation , Full-time Engineer | Japan
Apr 2026 – present
4 months |
| LY Corporation R&D , Part-time Researcher <ul style="list-style-type: none">Improved phonemic and prosodic annotation models for text-to-speech; the work was accepted to Interspeech 2025.Researched neural vocoders for high-quality speech generation from self-supervised speech features; the work was accepted to ICASSP 2026. | Japan
Dec 2024 – Mar 2026
1 year 4 months |
| NTT Communication Science Laboratories , Research Intern <ul style="list-style-type: none">Researched post-refinement of ASR hypotheses for joint multi-channel distant speech recognition. | Japan
Feb 2025 – Mar 2025
1 month |
| LY Corporation R&D , Research Intern <ul style="list-style-type: none">Improved phonemic and prosodic annotation models for text-to-speech and related subtasks. | Japan
Aug 2024 – Oct 2024
3 months |
| The University of Tokyo , Research Intern <ul style="list-style-type: none">Researched environmental sound synthesis from visual onomatopoeia; the work was accepted to ICASSP 2023. | Japan
June 2022 – July 2022
2 months |

Education

- | | |
|--|------------------------------|
| M.S. Nara Institute of Science and Technology , Information Science <ul style="list-style-type: none">Graduate School of Science and Technology, Master's Course.Intelligent Robot Dialogue Laboratory (RIKEN), supervised by Prof. Koichiro Yoshino.Received the NAIST Best Student Award. | Japan
Apr 2024 – Mar 2026 |
| National Institute of Technology, Tokuyama College , Computer Science and Electronic Engineering <ul style="list-style-type: none">Advanced Course.Miyazaki Laboratory, supervised by Prof. Ryoichi Miyazaki. | Japan
Apr 2022 – Mar 2024 |
| National Institute of Technology, Tokuyama College , Computer Science and Electronic Engineering <ul style="list-style-type: none">Regular Course.Miyazaki Laboratory, supervised by Prof. Ryoichi Miyazaki. | Japan
Apr 2017 – Mar 2022 |

Selected Publications

- | | |
|--|---------------|
| Wave-Trainer-Fit: Neural Vocoder with Trainable Prior and Fixed-Point Iteration towards High-Quality Speech Generation from SSL Features
Hien Ohnaka, Yuma Shirahata, Masaya Kawamura
(IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)) | Accepted 2026 |
|--|---------------|

Grapheme-Coherent Phonemic and Prosodic Annotation of Speech by Implicit and Explicit Grapheme Conditioning Hien Ohnaka, Yuma Shirahata, Byeongseon Park, Ryuichi Yamamoto arxiv.org/abs/2506.04527 (Interspeech)	Aug 2025
Unsupervised Speech Enhancement with Spectral Kurtosis and Double Deep Priors Hien Ohnaka, Ryoichi Miyazaki doi.org/10.1250/ast.e24.66 (Acoustical Science and Technology)	Feb 2025
Visual Onoma-to-Wave: Environmental Sound Synthesis from Visual Onomatopoeias and Sound-Source Images Hien Ohnaka, Shinnosuke Takamichi, Keisuke Imoto, Yuki Okamoto, Kazuki Fujii, Hiroshi Saruwatari doi.org/10.1109/ICASSP49357.2023.10096517 (IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP))	June 2023

Awards

Best Student Award Awarded by Nara Institute of Science and Technology. Announcement	Mar 2026
Committee Special Award Awarded by the Association for Natural Language Processing for work on jointly predicting response timing and short responses for real-time spoken dialogue systems. Award details	Mar 2025
IEEE SPS Tokyo Joint Chapter Student Award Awarded for Visual Onoma-to-Wave. Award details	Dec 2024

Domestic Presentations (Non-peer-reviewed)

音声の意味的類似度に基づいた対照学習によるリアルタイム応答選択 大中 緋慧, 大西 一誉, 吉野 幸一郎 (言語処理学会第 32 回年次大会)	Mar 2026
Voice Activity Projection とガンマ分布ハザード関数を用いて対話の間を考慮したターンシフトモデル 大西 一誉, 大中 緋慧, 吉野 幸一郎 (言語処理学会第 32 回年次大会)	Mar 2026
リアルタイム音声対話システムのための応答タイミングと短文応答の同時予測 大中 緋慧, 河野 誠也, 大西 一誉, 吉野 幸一郎 www.anlp.jp/proceedings/annual_meeting/2025/pdf_dir/P10-4.pdf (言語処理学会第 31 回年次大会)	Mar 2025
話者特性に基づくターンテイキング速度の分析 大西 一誉, 大中 緋慧, 吉野 幸一郎 www.anlp.jp/proceedings/annual_meeting/2025/pdf_dir/D8-1.pdf (言語処理学会第 31 回年次大会)	Mar 2025
情報案内・傾聴対話における対話行為とターン交替時間の関係分析 大中 緋慧, 河野 誠也, 大西 一誉, 吉野 幸一郎 doi.org/10.11517/jsaislud.102.0_96 (人工知能学会研究会資料 言語・音声理解と対話処理研究会)	Nov 2024
F0 に基づいて伸縮された画像文字からの音声合成 大中 緋慧, 宮崎 亮一, 高道 慎之介 speakerdeck.com/neh0615/f0niji-duiteshen-suo-saretahua-xiang-wen-zi-karanoyin-sheng-he-cheng (日本音響学会講演論文集)	Mar 2024

事前学習不要な DNN 音声強調の反復処理に基づく性能改善 中田 優翔, 広政 遼汰, 大中 緋慧, 宮崎 亮一 (日本音響学会講演論文集)	Mar 2024
クリーン不要な DNN 音声雑音除去に用いる目的信号の分析 広政 遼汰, 大中 緋慧, 宮崎 亮一 (日本音響学会講演論文集)	Mar 2024
過学習の抑制に基づく事前学習不要な DNN 音声強調 中田 優翔, 広政 遼汰, 大中 緋慧, 宮崎 亮一 (第 26 回関西支部若手研究者交流研究発表会)	Dec 2023
Double-DIP に基づく教師なし雑音除去の分析と性能改善 大中 緋慧, 宮崎 亮一 (日本音響学会講演論文集)	Mar 2023
Visual onoma-to-wave : 音源画像と画像オノマトペを利用した環境音合成の提案 大中 緋慧, 高道 慎之介, 井本 桂右, 岡本 悠希, 藤井 一貴, 猿渡 洋 (電子情報通信学会 音声研究会)	Feb 2023
クリーン音声を必要としない時間領域での DNN 音声強調 広政 遼汰, 大中 緋慧, 宮崎 亮一 (第 25 回関西支部若手研究者交流研究発表会)	Nov 2022
Double-DIP に基づく単一雑音混入音声のみを用いた雑音除去の検討 大中 緋慧, 藤村 拓弥, 宮崎 亮一 (電子情報通信学会 応用音響研究会)	May 2022
深層ニューラルネットワークにおける学習過程の信号分布に着目した雑音抑圧の検討 大中 緋慧, 藤村 拓弥, 宮崎 亮一 (第 24 回関西支部若手研究者交流研究発表会)	Dec 2021

Research Projects

Wave-Trainer-Fit A neural vocoder with trainable priors and fixed-point iteration for high-quality speech generation from self-supervised speech features. • Accepted to ICASSP 2026.	2025 – 2026
Unsupervised Speech Enhancement with Spectral Kurtosis and Double Deep Priors An unsupervised speech enhancement method combining spectral kurtosis and deep priors. • Published in Acoustical Science and Technology.	2024 – 2025
Visual Onoma-to-Wave Environmental sound synthesis from visual onomatopoeias and sound-source images. • Accepted to ICASSP 2023.	2022 – 2023